

Primary prevention of stroke by healthy lifestyle



Background The combination of healthy lifestyle factors is associated with lower risk of coronary heart disease, diabetes and total cardiovascular disease. Little is known about the impact of multiple lifestyle factors on risk of stroke. **Methods and results** We conducted a prospective cohort study among 43,685 men from Health Professionals Follow-up Study and 71,243 women from the Nurses' Health Study. Diet and other lifestyle factors were updated from self-reported questionnaires. We defined a low-risk lifestyle as not smoking, a body mass index $<25 \text{ kg/m}^2$, ≥ 30 minutes/day of moderate activity, consuming alcohol modestly (men: 5–30g; women: 5–15g alcohol/day), and scoring within the top 40% of a healthy diet score. We documented 1559 strokes (853 ischemic, 278 hemorrhagic) among women and 994 strokes (600 ischemic, 161 hemorrhagic) among men during follow-up. Women with all five low-risk factors had a relative risk of 0.21 (95%CI: 0.12, 0.36) for total and 0.19 (95%CI: 0.09, 0.40) for ischemic stroke, compared to women who had none of these factors. Among men, the relative risks were 0.31 (95%CI: 0.19, 0.53) for total and 0.20 (95%CI: 0.10, 0.42) for ischemic stroke for the same comparison. Among the women, 47% (95%CI: 18%, 69%) of total and 54% (95%CI: 15%, 78%) of ischemic stroke cases were attributable to lack of adherence to a low-risk lifestyle; among the men, 35% (95%CI: 7%, 58%) of total and 52% (95%CI: 19%, 75%) of ischemic stroke may have been prevented. **Conclusions** A low-risk lifestyle that is associated with a reduced risk of multiple chronic diseases may also be beneficial in the prevention of stroke, especially ischemic stroke.